

VADSO, Norway

WMO: 010880

Lat: 70.065N

Lon: 29.835E

Elev: 39

StdP: 100.86

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.8	-13.8	-18.3	0.8	-15.3	-16.2	0.9	-13.3	17.3	-2.1	15.6	-2.8	5.8	280	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	19.8	15.0	17.7	13.7	15.9	12.7	15.5	19.0	14.1	16.9	13.0	15.4	3.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.8	17.3	12.6	9.1	15.9	11.5	8.5	14.4	43.5	19.0	39.8	17.0	36.7	15.5	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.6	11.1	DB	-19.3	24.5	2.4	2.8	-21.1	26.5	-22.5	28.2	-23.9	29.7	-25.6	31.7	(4)
(5)			WB	-18.6	17.0	2.9	1.9	-20.7	18.4	-22.3	19.5	-23.9	20.5	-26.0	21.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-5.8	-6.5	-4.2	-0.9	4.1	7.9	11.1	10.3	7.0	2.0	-2.3	-3.5	(6)	
	DBStd	6.94	3.69	4.37	3.63	3.22	3.22	3.27	3.12	2.80	2.84	3.29	3.45	3.27	(7)	
	HDD10.0	3172	490	462	441	327	187	82	22	29	97	249	369	419	(8)	
	HDD18.3	6093	748	695	699	577	443	313	227	248	339	507	619	677	(9)	
	CDD10.0	122	0	0	0	0	2	18	55	39	7	0	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)	
	CDH23.3	11	0	0	0	0	0	0	8	2	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.7	6.6	6.7	6.7	5.5	5.3	5.5	4.8	4.3	4.8	5.4	6.0	6.4	(14)	
(15) Precipitation	PrecAvg	554	46	42	36	29	32	46	59	61	54	65	42	44	(15)	
	PrecMax	673	74	61	63	56	59	95	131	153	87	108	72	66	(16)	
	PrecMin	461	24	21	10	11	17	7	12	24	31	24	12	17	(17)	
	PrecStd	56	11	10	13	11	13	24	33	30	14	20	17	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	3.2	4.2	8.6	16.2	20.3	24.4	21.8	16.4	10.2	6.0	4.0	(19)	
		MCWB	2.7	2.0	2.1	5.6	10.6	15.0	17.4	16.8	13.0	8.1	4.9	2.8	(20)	
	2%	DB	2.2	2.1	3.0	6.0	13.3	17.7	20.8	18.4	14.2	8.8	4.2	3.0	(21)	
		MCWB	0.8	0.8	1.4	3.8	8.8	12.9	15.5	14.3	11.4	7.4	3.2	2.0	(22)	
	5%	DB	0.9	1.0	2.1	4.4	11.1	15.2	18.4	16.5	12.8	7.6	3.1	2.2	(23)	
		MCWB	-0.2	-0.1	0.6	2.8	7.5	11.4	14.4	13.3	10.6	6.5	2.1	1.0	(24)	
	10%	DB	-0.3	-0.3	1.0	3.4	9.1	13.0	16.2	15.0	11.4	6.6	2.2	1.2	(25)	
		MCWB	-1.2	-1.3	-0.4	2.1	6.3	9.8	13.3	12.5	9.4	5.5	1.2	0.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.2	2.4	5.6	11.1	15.8	17.9	17.1	13.5	8.3	5.0	2.9	(27)	
		MCDB	3.7	3.2	3.9	8.0	14.8	19.6	23.4	20.7	15.9	9.5	5.7	3.8	(28)	
	2%	WB	1.0	1.0	1.6	4.0	9.0	13.3	16.0	15.0	11.8	7.4	3.3	2.1	(29)	
		MCDB	1.9	2.0	2.9	5.5	12.9	16.8	20.1	17.8	13.7	8.5	4.1	3.0	(30)	
	5%	WB	-0.2	-0.1	0.7	3.0	7.8	11.7	14.7	13.7	10.7	6.5	2.2	1.2	(31)	
		MCDB	0.7	1.0	1.9	4.2	10.8	15.0	17.8	15.9	12.5	7.4	3.0	2.0	(32)	
	10%	WB	-1.2	-1.3	-0.2	2.3	6.5	10.1	13.3	12.7	9.8	5.5	1.3	0.2	(33)	
		MCDB	-0.3	-0.4	0.9	3.3	8.9	12.7	16.0	14.7	11.2	6.4	2.1	1.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.0	4.8	5.3	5.8	5.6	5.9	6.1	5.3	3.8	4.3	4.5	(35)	
		MCDBR	5.6	4.8	4.2	5.1	9.5	9.2	10.3	8.8	7.1	3.8	3.9	3.9	(36)	
	5% WB	MCWBR	5.1	4.5	3.7	3.9	5.9	5.6	5.7	5.4	5.0	3.3	3.7	3.8	(37)	
		MCWBR	5.6	4.8	4.1	4.6	9.0	9.0	9.5	8.0	6.6	3.5	3.9	4.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.263	0.312	0.338	0.360	0.352	0.368	0.351	0.317	0.275	N/A	N/A	(40)	
		taud	N/A	2.038	2.121	2.192	2.317	2.441	2.430	2.457	2.497	2.180	N/A	N/A	(41)	
	Ebn at Noon		0	430	669	784	817	841	809	778	705	419	0	0	(42)	
		Edh at Noon	0	41	79	101	102	93	92	79	57	35	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.29	1.47	3.16	4.29	4.76	4.33	2.95	1.56	0.43	0.00	0.00	0.00	(44)	
	RadStd	0.00	0.03	0.10	0.24	0.46	0.36	0.42	0.29	0.14	0.05	0.00	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page